

$$P_N = \text{diag}(\underbrace{1, \dots, 1}_N, \underbrace{0, \dots, 0}_{30-N})$$

$$\hat{X}_{g,N} = \text{IDCT}(P_N \text{DCT}(X_g))$$

$$R_{g,N} = X_g - \hat{X}_{g,N}, \quad N = 2$$